

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Pharm. Examination University Theory Examination Nov-Dec 2015 PH115 Pharmaceutical Engineering-I

Date: 23/12/2015 , Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1 Attempt all questions. Each question is of one mark. 20
1. Laminar flow of a Newtonian fluid exist, when the Reynolds number is
[A] 1500
[B] 2800
[C] 4000
[D] All of the above
 2. An instrument which offers no obstruction to the flow and suitable for flow measurement is
[A] Orificemeter
[B] Venturimeter
[C] Rotameter
[D] All of the above
 3. The heat transfer which occurs at fluid is
[A] Conduction
[B] Convection
[C] Radiation
[D] All of the above
 4. Fourier's law is applicable when
[A] Mode of Heat Transfer is conduction + Radiation
[B] Mode of Heat transfer by conduction
[C] Mode of Heat transfer by convection
[D] Mode of Heat transfer by Radiation
 5. Viscosity of a fluid can be defined as
[A] Change in density of the fluid per unit temperature
[B] Flow resistance offered by the fluid

- [C]Flow velocity change
[D]All of the above
6. Which of the following is a tamper resistant packaging
[A]Blister Package
[B]Strip Package
[C]Aerosol Containers
[D]All of the above
7. Which one of the following metal used to improve corrosion resistance property of Carbon steel?
[A]Chromium
[B]Nickel
[C]Silicon
[D]Molybdenum
8. Which series of Stainless steel is widely used for Pharma equipment construction?
[A]Type 430
[B]Type 410
[C]Type 243
[D]Type 316
9. Treated soda lime glass is also known as
[A]Type I
[B]Type II
[C]Type III
[D]Type IV
10. Glass composition is basically of
[A] Alkali based
[B]Acidic based
[C] Silicon dioxide based
[D] (a) & (c)
11. A pipe with red colour is used for handling of
[A]Fire fluids
[B]Gas
[C]Semisolids
[D]None of the above
12. Body that radiates maximum possible quantity of energy is
[A]White body
[B]Black body
[C]Grey body
[D]None of the above

13. _____ solid has highest thermal conductivity.
- [A]Copper
 - [B]Aluminium
 - [C]Stainless steel
 - [D]Glass
14. Types of steam traps are
- [A]Convective & Conduction
 - [B]Electrostatic
 - [C]Valve actuated
 - [D]Mechanical & thermostatic
15. 1 poise = _____ centipoise
- [A] 10
 - [B]100
 - [C]1000
 - [D]None of the above
16. Convert 1 Calorie Heat into _____ Joule heat.
- [A]4.187
 - [B]252
 - [C]3.086
 - [D]1.055
17. Convert 50 Farenhit temperature into _____ degree celcius .
- [A]10
 - [B]20
 - [C]30
 - [D]50
18. 5 m/hr convert into 5m/sec
- [A]0.0014
 - [B]0.083
 - [C]0.01
 - [D]None of the above
19. Convert 14.70 PSI pressure into _____ Bar pressure.
- [A]760
 - [B]1.013
 - [C]2117
 - [D]10333
20. Convert 10 cm distance into _____ Inch distance
- [A]3.94
 - [B]5
 - [C]10
 - [D]2.54

SECTION – II

- Q 2 Attempt any FOUR of the following.**
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|-------------|--|-----------|
| A | Answer the following. (Each of 1 mark) | 05 |
| a) | Give the Bernoulli's theorem for total energy balance. | |
| b) | Define Stoichiometry. | |
| c) | Classify the types of fluids as per viscosity behaviour. | |
| d) | Define Ties substance with example. | |
| e) | Define steady state process. | |
| B a) | Give the various types of fluid flows exist in pipes. | 03 |
| b) | Differentiate Orifice meter with Venturimeter. | 02 |
| C | Briefly explain the Reynolds experiment for measurement of type of flow. | 05 |
| D | A salt solution originally contains 4% by weight sodium sulphate in water is evaporated to 7% by weight sodium sulphate . What % of water is evaporated? | 05 |
| E | Classify the various flow meters for measurement of fluid flow in pipe with suitable example. | 05 |
| F | Briefly describe the fundamental unit systems. | 05 |

SECTION – III

- Q 3 Attempt any FOUR of the following.**
- | | | |
|----------|--|-----------|
| A | Describe steam as an ideal heating medium. | 05 |
| B | Write a short note on different modes of heat transfer. | 05 |
| C | Classify the materials for Plant construction. Give the factors affecting the selection of material for plant construction. | 05 |
| D | Write short note on 'Color code for piping system in Industry'. | 05 |
| E | Describe construction working and advantage of belt conveyor with a neat sketch. | 05 |
| F | Calculate the heat conducted through a 0.2 m thick industrial furnace wall made of fireclay brick. Measurements made during steady state operation showed that the wall temperatures inside and outside the furnace are 1500 and 1100 K respectively. The length of the wall is 1.2 m and the height is 1 m. | 05 |
- Q 4 Attempt any FOUR of the following.**
- | | | |
|----------|---|-----------|
| A | Explain the terms Black body & Grey body. State and explain Stefan-Boltzmann's law of heat radiation. | 05 |
| B | State and explain Fourier's law of heat transmission with equation. | 05 |
| C | Write a note on Fick's law of mass transfer. | 05 |
| D | Enlist the types of packaging system. Describe glass as an ideal packaging material. | 05 |
| E | Write note on temper resistant packaging of pharmaceutical products. | 05 |
| F | Briefly describe the types and theory of corrosion. | 05 |